INCH-POUND

MS20663E
10 January 2001
SUPERSEDING
MS20663D
30 October 1999

MILITARY SPECIFICATION SHEET

BALL END, WIRE ROPE, SWAGING, DOUBLE SHANK

This specification is approved for use by all departments and agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation: MIL-DTL-781, QPL-781.

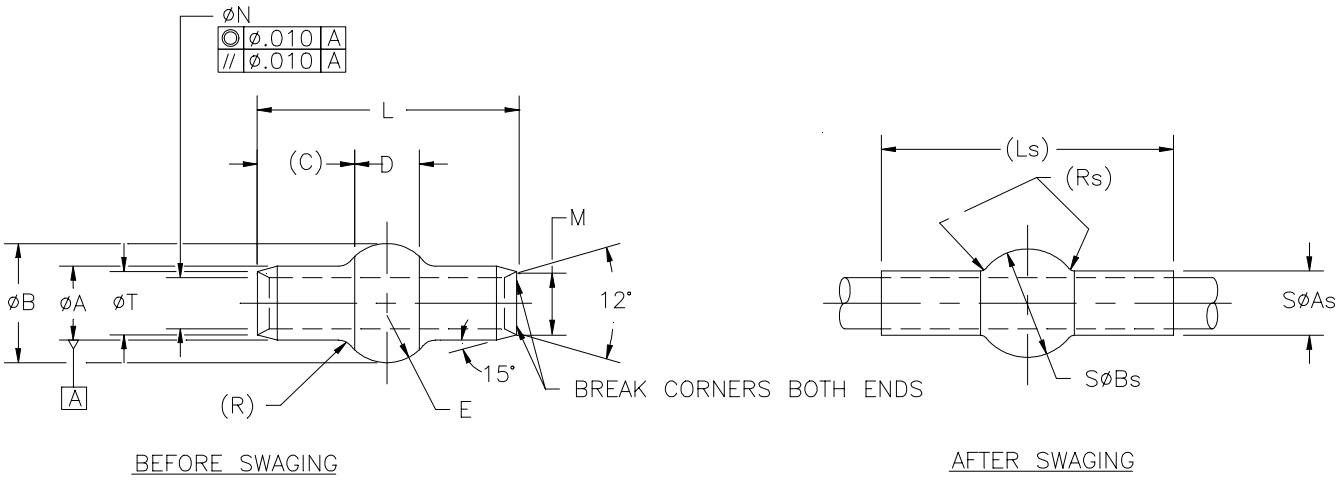

FIGURE 1. Ball end, wire rope, swaging, double shank.

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TABLE I. Part numbers and dimensions.

MS part number Corrosion resistant steel	Wire rope diameter		Minimum breaking strength lb	∅A		∅As		∅B		S∅Bs spherical		C REF
	Nominal	Minimum										
MS20663C2	1/16	.062	480	.127	+.000 -.004	.112	+.000 -.003	.207	+.000 -.004	.190	+.000 -.003	.125
MS20663C3	3/32	.093	920	.163		.143		.277		.253		.188
MS20663C4	1/8	.125	2,000	.218		.190		.345		.315		.250
MS20663C5	5/32	.156	2,800	.254		.222	+.000 -.004	.419		.379	+.000 -.004	.313
MS20663C6	3/16	.187	4,200	.293	+.000 -.005	.255	+.000 -.005	.487	+.000 -.005	.442	+.000 -.005	.375
MS20663C7	7/32	.218	5,600	.347		.302		.555		.505		.438
MS20663C8	1/4	.250	7,000	.401		.348		.624		.567		.500
MS20663C9	9/32	.281	8,000	.439		.382	+.000 -.007	.694		.632	+.000 -.007	.562
MS20663C10	5/16	.312	9,800	.475		.413		.763		.694		.625

TABLE I. Part numbers and dimensions - Continued.

MS part number Corrosion resistant steel	D reference		E radius reference	L ±.0075	Ls reference	M +.010 -.000	∅N		R radius reference	Rs radius reference	∅T +.000 -.010
MS20663C2	.114	+.000 -.003	.0825	.3620	.390	.079	.073	+.004 -.000	.028	.014	.109
MS20663C3	.152	+.000 -.004	.110	.5250	.578	.114	.104		.037	.019	.141
MS20663C4	.1895	+.000 -.005	.137	.6880	.765	.152	.139		.046	.023	.188
MS20663C5	.2270		.165	.8500	.953	.185	.169		.056	.028	.219
MS20663C6	.2640		.192	1.0120	1.140	.221	.201		.065	.033	.250
MS20663C7	.3025		.220	1.1750	1.328	1/	.233	+.005 -.000	.075	.038	.297
MS20663C8	.3395		.247	1.3370	1.515		.264		.084	.042	.344
MS20663C9	.3775		.275	1.4970	1.719		.297		.094	.046	.375
MS20663C10	.4145		.302	1.6640	1.875		.328		.103	.046	.406

1/ No tapered bellmouth on –7, -8, -9, -10.
2/ Reference dimensions are for design purposes only and are not an inspection requirement.

REQUIREMENTS:

1. Material: Material shall be in accordance with MIL-DTL-781.

2. Finish: Finish shall be in accordance with MIL-DTL-781.

3. Swage: Swage shall be in accordance with MIL-DTL-6117.

4. Tolerances: Unless otherwise specified, tolerances: decimals ± .010, angles ± 3°.

NOTES:

1. Dimensions are in inches.

2. Interpret drawing in accordance with ASME Y14.5M.

3. Remove all burrs and sharp edges. (See Procurement specification for definition.)

4. Interchangeability relationship: MS20663 and AN663 parts identified by the same dash number are universally, functionally and dimensionally interchangeable.

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5. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence.

6. Unless otherwise specified, issues of reference documents are those in effect at the time of solicitation.

7. MS20663 supersedes AN663.

8. Corrosion resistant steel parts can be universally substituted for carbon and alloy steel parts in accordance with table II. Carbon and alloy steel parts cannot be substituted for corrosion resistant steel parts.

TABLE II. Substitution table.

MS PART NUMBER	
Corrosion resistant steel	Carbon steel, cadmium plated
MS20663C2	MS20663-2
MS20663C3	MS20663-3
MS20663C4	MS20663-4
MS20663C5	MS20663-5
MS20663C6	MS20663-6
MS20663C7	MS20663-7
MS20663C8	MS20663-8
MS20663C9	MS20663-9
MS20663C10	MS20663-10

9. Carbon and alloy steel parts are inactive for new design.

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR4
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